

# LAKEFRONT WINDOW

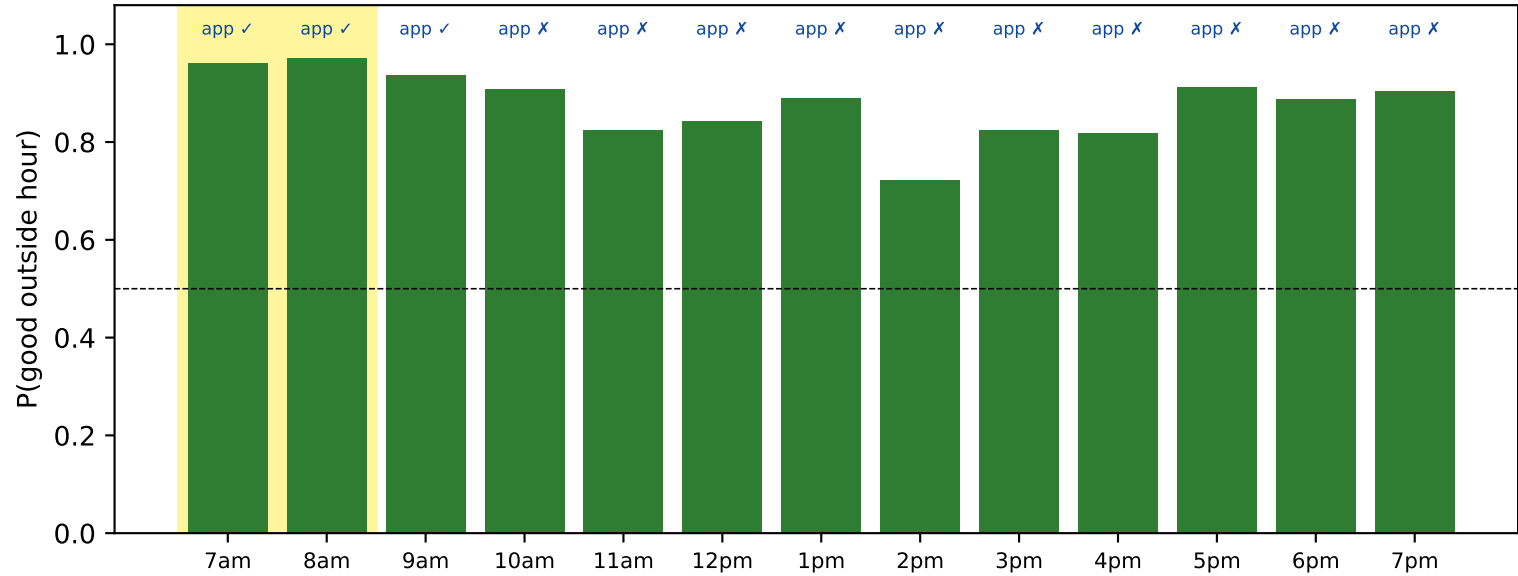
Built with PriorLabs-TabPFN

Oak Street Beach, Chicago · Tue Jul 1, 2025 · Replay of a holdout day

## Best 2 hours: 7am-9am

Chance the lakefront is a 'good outside hour' in that window: 97% · sunrise 5:18 AM, sunset 8:29 PM

Hour by hour: TabPFN-v2 lakefront estimate (hatched = not full daylight)



| Hour | Temp | Wind   | Rain | Lakefront P(good) | App says OK? | Actual at lakefront |
|------|------|--------|------|-------------------|--------------|---------------------|
| 7am  | 75°F | 8 mph  | 0%   | 96%               | yes          | good                |
| 8am  | 77°F | 9 mph  | 0%   | 97%               | yes          | good                |
| 9am  | 80°F | 10 mph | 0%   | 94%               | yes          | good                |
| 10am | 81°F | 12 mph | 0%   | 91%               | no           | good                |
| 11am | 82°F | 7 mph  | 0%   | 82%               | no           | good                |
| 12pm | 84°F | 9 mph  | 0%   | 84%               | no           | good                |
| 1pm  | 85°F | 9 mph  | 0%   | 89%               | no           | good                |
| 2pm  | 85°F | 9 mph  | 0%   | 72%               | no           | good                |
| 3pm  | 87°F | 9 mph  | 0%   | 82%               | no           | good                |
| 4pm  | 88°F | 7 mph  | 0%   | 82%               | no           | good                |
| 5pm  | 85°F | 13 mph | 0%   | 91%               | no           | good                |
| 6pm  | 84°F | 7 mph  | 0%   | 89%               | no           | good                |
| 7pm  | 84°F | 8 mph  | 0%   | 90%               | no           | good                |

What 'good' means: lakefront sensor reads 10-27 °C (50-81 °F), hourly gust under 9 m/s (~20 mph), no rain or snow.  
How: TabPFN-v2 (open-weight tabular model, run locally on CPU) learned how the Oak Street lakefront sensor differs from Midway airport conditions (spring/summer lake cooling, onshore wind) and applies that to the inputs.  
Holdout 2024-2026 (11,397 daylight-range hours, airport obs as input): TabPFN accuracy 91.4% vs 84.6% for 'trust the weather app as-is'; best-window hit rate 56% vs 51%. Real forecast error is NOT included in these numbers.  
Inputs: actual Midway airport observations (a perfect 'forecast'); model trained on 2015-2023 only.  
Data: Chicago Data Portal k7hf-8y75 · IEM ASOS archive (KMDW) · api.weather.gov. Not a safety tool: check beach and marine advisories. Built by Alfred, an AI agent, for Brooks Moore. MIT licensed.  
Now fold this card, leave the phone, and go.